

Jiangrui Yu

Ph.D. Candidate | Efficient and Secure AI Systems
Beijing, China | jiangrui.yu@stu.pku.edu.cn | Personal Website | Google Scholar

Education

Ph.D. Candidate in Integrated Circuit Science and Engineering School of Integrated Circuits and Institute for Artificial Intelligence, Peking University • Advisor: Prof. Meng Li	2024-Present
B.Sc. in Applied Physics School of Electronic Engineering and Computer Science, Peking University • Dual Degree: Computer Science and Technology	2020-2024

Research Interests

Focus: Efficient and Secure AI Systems, Software-Hardware Co-Design, and Privacy-Preserving Inference
Exploring: LLM and Omni-Modal Inference Infrastructure; Multimodal Generation and Post-Training

Research Experience

Ant Group , Intern Mentors: Mingzhe Zhang and Liang Kong • Topic: Compiler and cryptographic protocol design for private AI acceleration	Aug 2025 – Aug 2026
Peking University , Student Researcher – Beijing, China Institute for Artificial Intelligence and School of Integrated Circuits • Design efficient encrypted machine learning systems through cross-layer co-design spanning cryptographic algorithms, GPU software, and specialized hardware • Current work includes hybrid CKKS/TFHE evaluation, BatchPBS, automated HE kernel generation, and FHE acceleration	June 2023 – present

Publications

ROSETTA: Efficient and Accurate Privacy-Preserving LLM Decoding via Hybrid CKKS/TFHE Evaluation <u>Jiangrui Yu</u> , Baosheng Zhang, Liang Kong, Lin Ding, Yi Chen, Ye Yu, Mingzhe Zhang, Meng Li <i>ACM SIGSAC Conference on Computer and Communications Security (CCS)</i> .	2026
OptiPrime: Optimizing Private Inference through Protocol-Hardware Co-Design <u>Jiangrui Yu</u> , Ye Yu, Si Chen, Chenqi Lin, Wenxuan Zeng, Junfeng Fan, Mingyu Gao, Meng Li <i>IEEE/ACM International Symposium on Microarchitecture (MICRO)</i> . Distinguished Paper Award @ UTI 2026	2026
(Invited) PEFT: A Near-Memory Processing-Enabled Heterogeneous Accelerator for BatchPBS TFHE <u>Jiangrui Yu</u> , Yi Chen, Meng Li <i>IEEE International Symposium on Circuits and Systems (ISCAS)</i> .	2026
Breaking the Layer Barrier: Remodeling Private Transformer Inference with Hybrid CKKS and MPC Tianshi Xu*, Wen-jie Lu*, <u>Jiangrui Yu</u> *, Yi Chen, Chenqi Lin, Runsheng Wang, Meng Li <i>34th USENIX Security Symposium</i> . * Equal contribution.	2025
FlexHE: A Flexible Kernel Generation Framework for Homomorphic Encryption-Based Private Inference <u>Jiangrui Yu</u> , Wenxuan Zeng, Tianshi Xu, Renze Chen, Yun (Eric) Liang, Runsheng Wang, Ru Huang, Meng Li <i>IEEE/ACM International Conference on Computer-Aided Design (ICCAD)</i> .	2024
(Invited) FENIX: Flexible and Efficient Hybrid HE/MPC Acceleration with Near-Memory Processing Tengyu Zhang, Chenqi Lin, <u>Jiangrui Yu</u> , Yi Chen, Shuwen Deng, Meng Li <i>IEEE/ACM International Conference on Computer-Aided Design (ICCAD)</i> .	2025
Trinity: A General Purpose FHE Accelerator Xianglong Deng, Shengyu Fan, Zhicheng Hu, Zhuoyu Tian, Zihao Yang, <u>Jiangrui Yu</u> , Dingyuan Cao, Dan Meng, Rui Hou, Meng Li, Qian Lou, Mingzhe Zhang <i>57th IEEE/ACM International Symposium on Microarchitecture (MICRO)</i> .	2024

Patents

An Efficient Homomorphic Computation Method for Privacy-Preserving Inference in Neural Networks	2026
Meng Li, <u>Jiangrui Yu</u> , Ye Yu, Wenxuan Zeng, Runsheng Wang	
China Patent Application, CN121418079A.	

Research and Competition Awards

Noteworthy Reviewer Recognition (33/222 AEC members), USENIX Security	2026
1st Place, DAC System Design Contest, DAC 2025	2025

University Honors

Doctoral Dean's Scholarship (Excellence Award), Institute for Artificial Intelligence, Peking University	2025
Outstanding Research Award, Peking University	2025
Huawei Scholarship (Third Prize)	2025
DACS Industry Contribution Award, Peking University	2024
Outstanding Graduate, Peking University and Beijing	2024
National Scholarship (Top 0.2%)	2023
Merit Student	2021-2023
Canon Scholarship	2022
Lihuirong Scholarship	2021
Bronze Medal, 35th Chinese Physics Olympiad (CPhO), as a second-year high school student	2018

Academic Service

Artifact Evaluation Committee Member (Full-term) – USENIX Security	2026
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Campus Leadership

Class Monitor – Peking University	Sept 2020 – July 2024
Deputy Director, Organizational Development Center, PKUICSS – Peking University	Sept 2022 – July 2023
Freshman Counselor – Peking University	Sept 2021